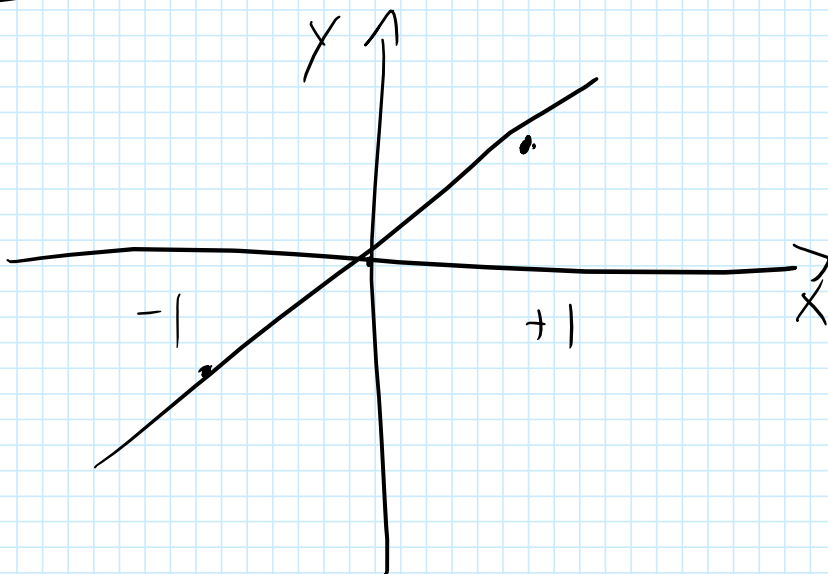


RETTA

$$Y = bX + c$$

$$\begin{matrix} x & y \\ (1, 2) \end{matrix}$$



IMPLICITA

$n \in \mathbb{N}, \mathbb{Z}, \mathbb{R}$

$$ax + by + c = 0$$

ESPLICITA

$$y = mx + q$$

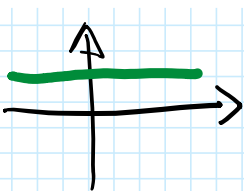
$$\frac{by}{b} = \frac{-ax}{b} - \frac{c}{b}$$

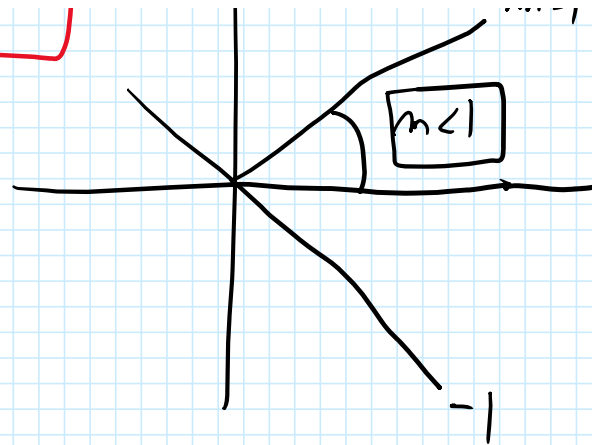
$$\begin{aligned} m &= \frac{-a}{b} \\ +q &= -\frac{c}{b} \end{aligned}$$

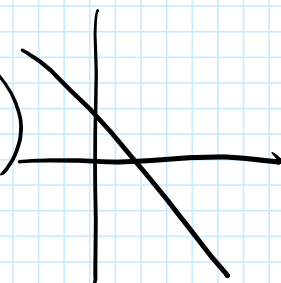
$m > 0$ ($y = 2x + 1$)

$m = 0$ ($y = 2$)

$m = 1$

$$m = 0 \quad (y = 3)$$


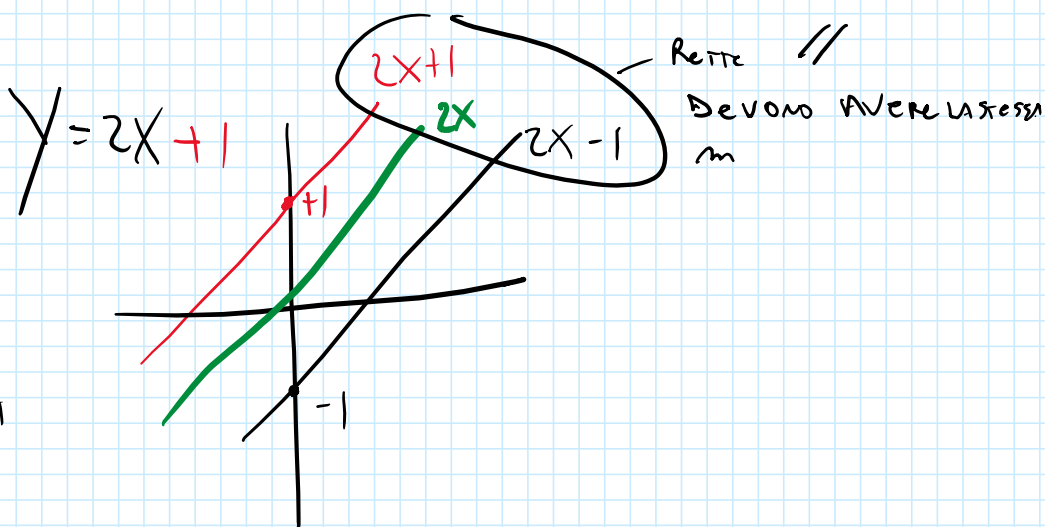


$$m < 0 \quad (y = -2x + 1)$$


$$q = 0 \quad (x = 0)$$

↓

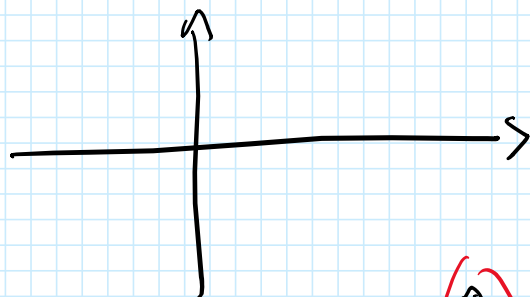
RETTE PASSA



RETTE VERTICALE

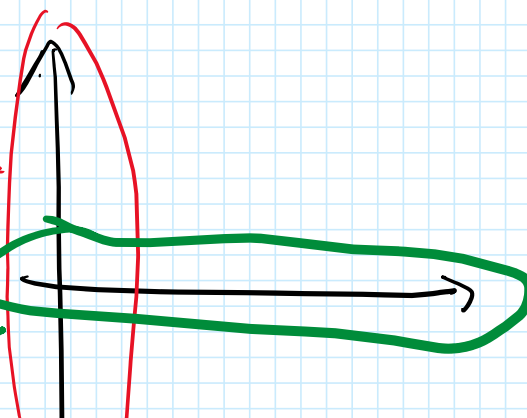
$$X = k_1$$

$$y = IND$$

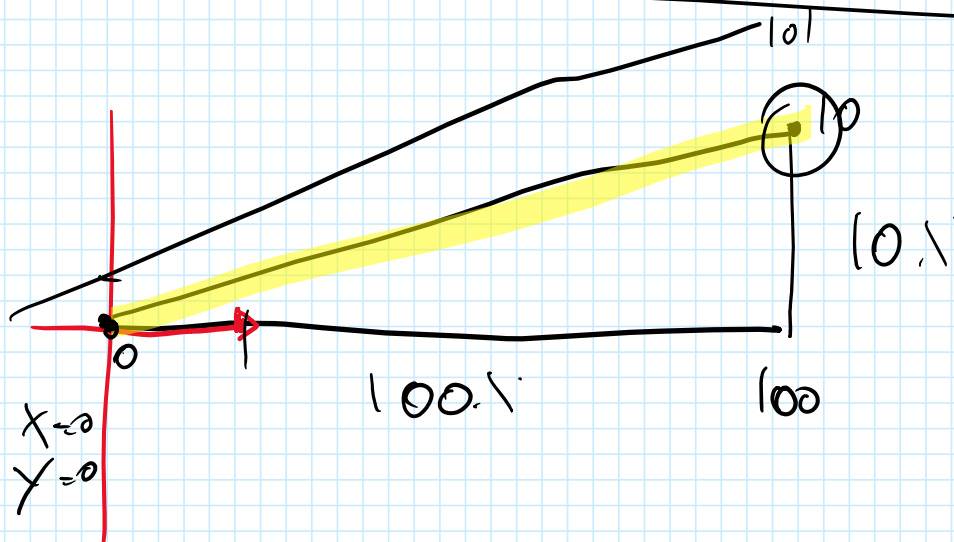


$$X = 0 \leadsto \text{Asse ORDINATE } (y)$$

$$Y = 0 \leadsto \text{Asse ASCISSE } (x)$$

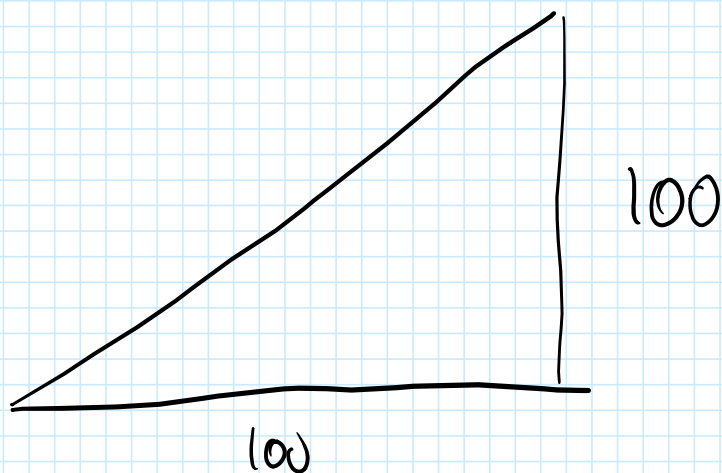


RETTA GENERICA $a \neq 0$
 $b \neq 0$



$$m = \frac{10}{100} = \frac{1}{10}$$

$$y = \frac{1}{10}x$$

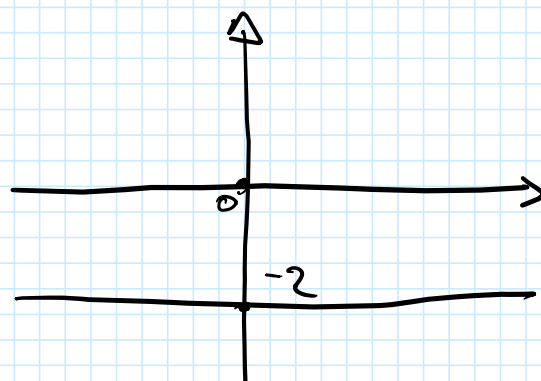


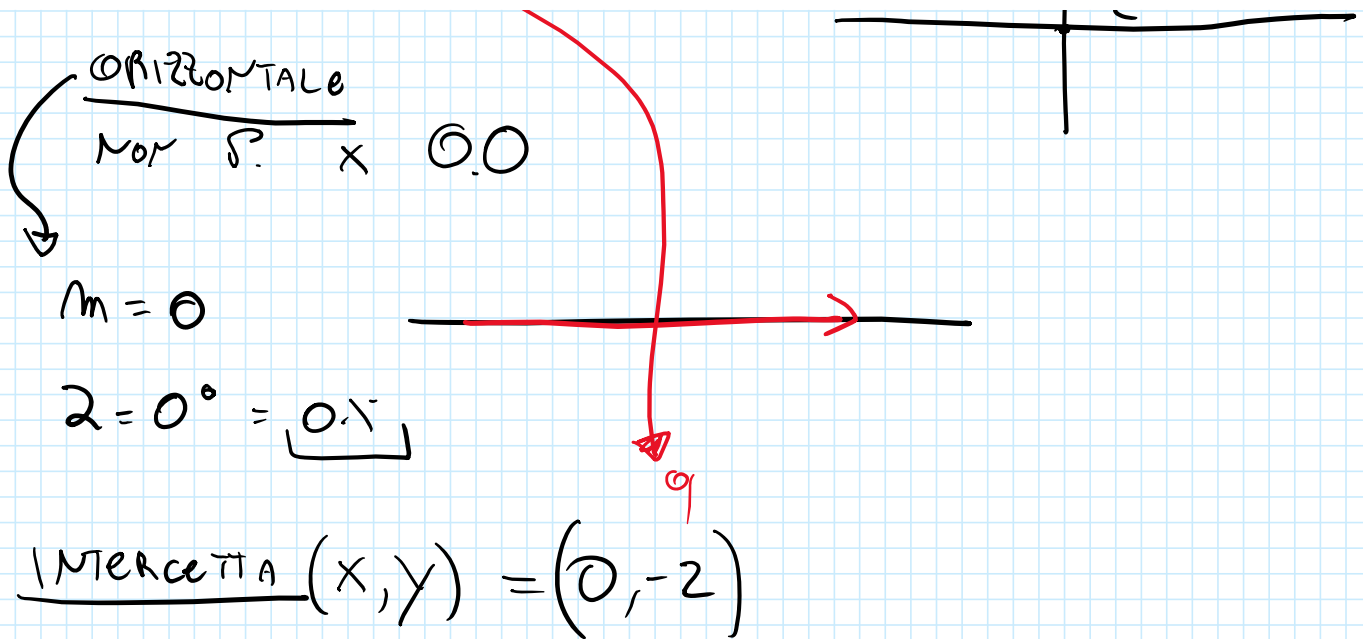
$$m = \frac{100}{100} = 1$$

2) $2y + 4 = 0$ IMPLICITA

$y = -2$ ESPLICITA

ORIZZONTALE





B) $y = -2x$ ESPlicita

$y = mx$

$+2x + y = 0$ IMPLICITA

x	y
0	0

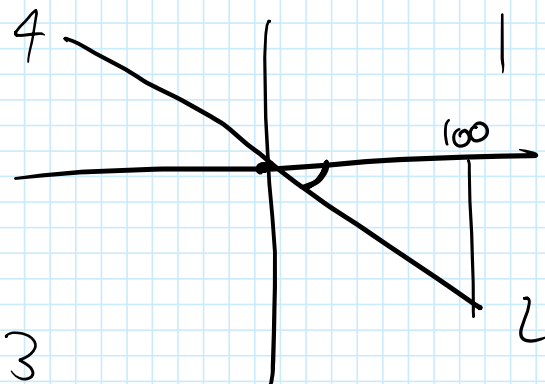
PASSA x 0.0

Generica

Coeff. ANGOLARE

$m = -2$

X	Y
100	-200



$m = \frac{y}{x} = \frac{-200}{100}$

$m = -2$

$-200x$

3

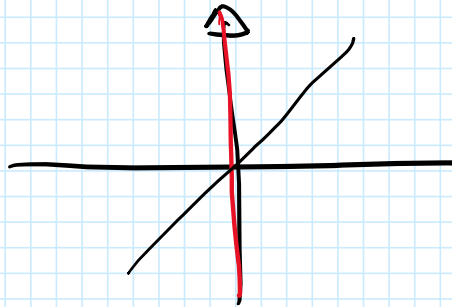
|

L

-200x

$$y = -2x$$

INTERCETTA



$$\begin{cases} X=0 \\ Y=-2X \end{cases}$$

ORD

$$\begin{cases} X=0 \\ Y=-2(0) \end{cases}$$

$$\begin{cases} X=0 \\ Y=0 \end{cases} \rightarrow \text{INTERCETTA}$$

(0,0)

$$5x - 20 = 0$$

$$5x = \frac{20}{5}$$

 $\forall y, x=4$

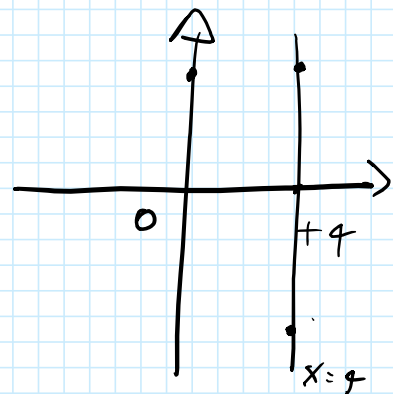
$$x = 4$$

VERTICALE

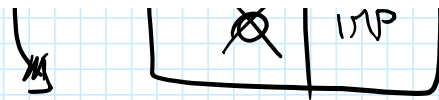
$$\text{IMPLICITA} \\ (ax + by + c = 0)$$

$$\text{ESPlicita} \\ (y = mx + q)$$

x	y
x	imp



VERTICALE



$$m=5$$

$$0y = 5x - 20$$

$$\text{Intercept} \begin{cases} X=0 \\ X=4 \end{cases} \text{ IMP}$$

d) $x + 3y - 6 = 0$ IMPLICITA

$$3y + x = 6$$

$$\frac{3y}{3} = \frac{-x + 6}{3}$$

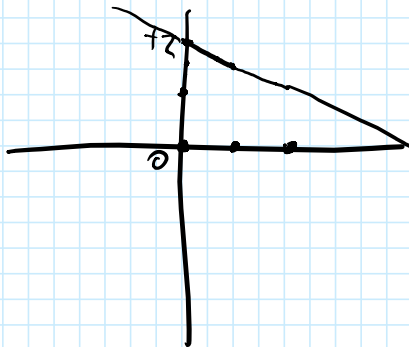
$$y = -\frac{1}{3}x + 2$$
 ESPlicita

X	Y
0	2
1	$\frac{5}{3}$
2	$\frac{4}{3}$

Coeff. ang

$$m = -\frac{1}{3}$$

$$\text{Intercept} (0, 2)$$



Pendenza decrescente

Pendenza decrescente



$$m = \frac{Y}{X} = \frac{-15}{100} = -0,15$$

$m \rightarrow$ Coef. ANGOLARE = pendenza

$$\begin{cases} m > 0 \rightarrow \text{RETTA CRESCENTE} \\ m = 0 \rightarrow \text{ORIZZ} \\ m < 0 \rightarrow \text{RETTA DECRESCENTE} \end{cases}$$

Se due rette HANNO LA STESSA m (m), sono //

$q \rightarrow$ INTERCETTA ALL'ORDINATA, PUNTO DI INCONTRO
CON ASSE Y

ASSE X $\rightarrow y = 0$

ASSE Y $\rightarrow x = 0$